Universal identities for Euclidean mechanical Lie brackets

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Abstract

We prove that Euclidean kinetic energy and arbitrary smooth potentials, considered in all dimensions, satisfy no universal Lie identities beyond those of the free polynomially graded Lie algebra of classical mechanics. Polynomial potentials already suffice to separate every nonzero element. The proof reduces positive momentum degree to degree zero: a sufficiently large binary-tree potential distinguishes words in degree one, while in higher degree a free Lie projection and a retraction recover a nonzero multiple of the original element. Symmetrization in a free pre-Lie algebra and finite Taylor jets then separate the potential part. This answers the Euclidean mechanical conjecture of McLachlan and Murua. Exact computations through order eighteen provide supporting finite verification.

1 Introduction

For a mechanical Hamiltonian

$$H(q,p) = T(p) + V(q), \qquad T(p) = \tfrac{1}{2}|p|^2,$$

the Poisson brackets of kinetic and potential energy determine the order conditions of splitting integrators. Quadratic dependence on momentum forces identities absent from an unrestricted free Lie algebra. For example, three successive brackets with a potential annihilate the kinetic energy. The question is whether a constant Euclidean kinetic metric imposes further identities that hold for every potential and every dimension.

McLachlan and Murua [1] formulated this question using the free polynomially graded Lie algebra of classical mechanics. Their structural theorem identifies its potential part with a free commutative nonassociative algebra, and its positive-degree part with a free Lie algebra on explicit generators. They also established faithfulness on the potential part by relating it to rooted trees and pre-Lie symmetrization. The corresponding assertion for all momentum degrees remained their Conjecture 1. The survey of Blanes, Casas and Murua [2, §3.1] retains the conjectural equality of the Euclidean and general mechanical order-condition reductions and reports prior checks through order twenty. Finite-order rank calculations alone cannot establish this equality in all orders.

Here is the precise free object and the statement we prove. Work over $\mathbb{R}$. Let $\mathfrak{P}$ be the class of Lie algebras $L = \bigoplus_{m\geq 0} L_m$ with

$$[L_n, L_m] \subseteq L_{n+m-1} \quad (n+m>0), \qquad [L_0, L_0] = 0. \tag{1}$$

Let F be free in this class on A of degree two and B of degree zero; every degree-preserving assignment of these generators extends uniquely to a homomorphism. For $d \geq 1$ and a smooth real potential V on $\mathbb{R}^d$, write

$$\Phi_{d,V} : F \longrightarrow C^\infty_{\mathrm{pol}}(\mathbb{R}^{2d}), \qquad A \longmapsto \tfrac{1}{2}|p|^2, \qquad B \longmapsto V(q).$$

The target consists of smooth functions polynomial in p, graded by homogeneous momentum degree, with the convention

$$\{f, h\} = \sum_{i=1}^{d} (f_{q_i} h_{p_i} - f_{p_i} h_{q_i}). \tag{2}$$

Theorem 1.1. *The Euclidean mechanical realizations have zero joint kernel:*

$$\bigcap_{d \geq 1} \bigcap_{V \in C^\infty(\mathbb{R}^d;\mathbb{R})} \ker \Phi_{d,V} = \{0\}. \tag{3}$$

The intersection is already zero when the potentials are restricted to real polynomials.

Thus every nonzero finite Lie expression is detected by some dimension and some polynomial potential. The quantifiers in (3) are those of the Euclidean conjecture and AIM problem 519; injectivity for an individual fixed potential is not asserted.

The new step is an algebraic degree-lowering argument. A putative universal identity belongs to a graded ideal. Bracketing with a suitable modified potential lowers its momentum degree without making it zero. The probe may have a much larger order than the expression being tested, which is permitted by the universal question. In degree one a large binary tree makes the ordered sequence of multiplications recoverable. In higher degree the probe acts, after discarding generators with larger tree labels, like replacing one independent kinetic letter by a fresh letter. Sending that letter back recovers the original expression multiplied by its kinetic-letter count. Every nonzero graded ideal must consequently meet the potential part, where faithfulness closes the proof.

Section 2 proves this reduction from the established free decomposition. Section 3 gives explicit tree and Taylor-jet arguments for the potential part and completes the theorem. Appendix B describes the finite computations. All Lie elements and tree combinations in the paper are finite sums; no completed Lie algebra is used.

2 Lowering momentum degree

Put $X = F_0$ and define

$$x * s = [s, [x, A]] \qquad (x, s \in X). \tag{4}$$

Jacobi and $[X, X] = 0$ show that this product is commutative. Theorem 1 of McLachlan and Murua [1], together with its Hall-basis proof in §2.3, gives the vector-space decomposition

$$F = X \oplus \mathfrak{g}, \qquad \mathfrak{g} = \bigoplus_{m \geq 1} F_m = \mathrm{Lie}_{\mathrm{free}}(A, \{y_s : s \in \mathcal{B}\}), \qquad y_s = [s, A]. \tag{5}$$

Here $\mathcal{B}$ is the basis of all unordered full binary products in B. With (4), X is the free *nonunital* commutative nonassociative algebra on B. The specified generators in (5), rather than abstract freeness alone, will be used below. If $|s|$ denotes the number of leaves, then

$$|s| \geq 1, \qquad |s * t| = |s| + |t|, \qquad [x, A] = y_x, \quad [x, y_s] = x * s.$$

For general $x \in X$, the notation $y_x = [x, A]$ is extended linearly.

Count the occurrences of A as an independent generator of $\mathfrak{g}$; the letters inside a label y_s are not counted. A Lie monomial with k such occurrences has momentum degree $k+1$, by (1). Freeness makes this a second grading, and hence

$$F_1 = \mathrm{Lie}_{\mathrm{free}}(\{y_s\}), \qquad F_m = \{\text{elements of } \mathfrak{g} \text{ homogeneous of independent } A \text{ count } m-1\}. \tag{6}$$

We use the injective associative realization of a free Lie algebra as commutator polynomials in its generators. An algebraic justification is given in Appendix A.

Lemma 2.1. *For every $0 \neq u \in F_1$, there is $x \in \mathcal{B}$ such that $[x, u] \neq 0$.*

Proof. Expand u as a nonzero finite associative polynomial $\sum_w c_w w$ in words $w = y_{s_1} \cdots y_{s_r}$. Choose

$$M > \max_{c_w \neq 0} \sum_{i=1}^{r} |s_i|, \qquad x \in \mathcal{B}, \quad |x| = M.$$

Such a tree exists for every positive M, for instance a comb. On X, $\mathrm{ad}_{y_s} = -L_s$, where $L_s(z) = s * z$. Thus the action of a word is

$$w(x) = (-1)^r s_1 * (s_2 * (\cdots * (s_r * x) \cdots)). \tag{7}$$

These output trees distinguish the words. At the root, the branch containing x has at least M leaves and the other branch has fewer than M. Removing the latter recovers s_1. Repeat to recover $s_2, \ldots, s_r$, stopping when the remaining tree has exactly M leaves. Every proper subtree of x has fewer leaves, so the stopping rule is unambiguous. Distinct associative words therefore have distinct outputs in the basis $\mathcal{B}$. Their coefficients, including the signs in (7), cannot cancel. Hence $[u, x] \neq 0$, and antisymmetry proves the lemma. □

For higher momentum degree we must control the additional potential terms created by this action. Fix $M \geq 1$ and set

$$X_{>M} = \mathrm{span}\{s \in \mathcal{B} : |s| > M\}, \qquad J_{>M} = \text{the Lie ideal of } \mathfrak{g} \text{ generated by } y_s \text{ with } |s| > M,$$

and $N_{>M} = X_{>M} \oplus J_{>M} \subset F$.

Lemma 2.2. *The subspace $N_{>M}$ satisfies*

$$[X_{>M}, \mathfrak{g}] \subseteq N_{>M}, \qquad [N_{>M}, \mathfrak{g}] \subseteq N_{>M}.$$

Proof. Induct on the length of a bracket expression w in the free generators of $\mathfrak{g}$, uniformly for all $z \in X_{>M}$. For a generator, $[z, A] = y_z \in J_{>M}$ and $[z, y_s] = z * s \in X_{>M}$, since $|s| \geq 1$. If $w = [v_1, v_2]$, Jacobi gives

$$[z, w] = [[z, v_1], v_2] + [v_1, [z, v_2]].$$

The induction hypothesis splits each $[z, v_i]$ into an $X_{>M}$ term and a $J_{>M}$ term. The former can be bracketed with the shorter expression v_j using the same hypothesis; the latter remains in $J_{>M}$ by its ideal property inside $\mathfrak{g}$. This proves the first inclusion. Adding $[J_{>M}, \mathfrak{g}] \subseteq J_{>M}$ gives the second. □

Lemma 2.3. *For every $m \geq 2$ and $0 \neq u \in F_m$, there is $x \in \mathcal{B}$ with $0 \neq [x, u] \in F_{m-1}$.*

Proof. Choose M strictly larger than every tree label in a finite free Lie expression for u, and choose $|x| = M$. If there are no such labels take any $M \geq 1$. In particular y_x is absent from u. Define a free associative derivation, restricted to $\mathfrak{g}$, by

$$\delta_x(A) = y_x, \qquad \delta_x(y_s) = 0 \quad (s \in \mathcal{B}).$$

On a word it replaces one occurrence of A by y_x and sums over all such positions. The commutator derivation rule shows that it preserves the free Lie subalgebra.

Regard both δ_x and ad_x as derivations $\mathfrak{g} \to F$. Their difference is zero on A and equals $x * s \in X_{>M}$ on y_s. Expanding a bracket by the derivation rule and applying Lemma 2.2 inductively yields

$$[x, u] - \delta_x u \in N_{>M}.$$

Both terms have momentum degree $m - 1 \geq 1$, so the difference actually lies in $N_{>M} \cap \mathfrak{g} = J_{>M}$.

Let π_M be the free Lie homomorphism that kills all y_s with $|s| > M$ and retains A and the other y_s. Its target is $\mathfrak{g}_{\leq M} = \mathrm{Lie}_{\mathrm{free}}(A, \{y_s : |s| \leq M\})$, identified with the subalgebra on these generators. Its kernel is exactly $J_{>M}$: the quotient $\mathfrak{g}/J_{>M}$ and $\mathfrak{g}_{\leq M}$ are isomorphic by inverse generator-preserving maps, by their universal properties. Consequently

$$\pi_M([x, u]) = \delta_x u. \tag{8}$$

The derivative uses only the old labels smaller than M and the new label of size M, so its image is retained.

Define $r_x : \mathfrak{g}_{\leq M} \to \mathfrak{g}$ by sending y_x to A and fixing every other generator, including A. For a bracket monomial without y_x, $r_x\delta_x$ multiplies it by its independent A count: this holds on each generator and the derivation rule adds the counts under a bracket. Equation (6) therefore gives

$$r_x \pi_M([x, u]) = (m - 1)u \neq 0. \tag{9}$$

Over $\mathbb{R}$ the scalar $m - 1$ is nonzero, proving the claim. These auxiliary maps are maps of ordinary free Lie algebras; no extension of r_x to F or grading preservation by r_x is required. □

Proposition 2.4. *Every nonzero graded Lie ideal of F has a nonzero intersection with X.*

Proof. If a nonzero graded ideal misses X, take its least occupied degree $m \geq 1$ and a nonzero homogeneous element u in that degree. Lemma 2.1 or Lemma 2.3 supplies a nonzero $[x, u]$ of degree $m - 1$ in the same ideal, a contradiction. □

3 Separating potentials and completing the proof

For completeness we make the degree-zero argument explicit, including the passage from formal trees to actual potentials. Let $\mathcal{U}$ be the vector space on finite nonempty unrooted simple trees. For two basis trees t, v, define $t \diamond v$ by joining one vertex of t to one vertex of v by a new edge and summing over all vertex pairs. Isomorphic outputs are counted with multiplicity. This product is commutative. Freeness of X defines a map

$$\psi : X \longrightarrow \mathcal{U}, \qquad \psi(B) = \text{one-vertex tree}, \qquad \psi(x * s) = \psi(x) \diamond \psi(s).$$

Lemma 3.1. *The map ψ is injective.*

Proof. Let $\mathcal{R}$ be the vector space on rooted trees, with $a \triangleright b$ obtained by attaching the root of b to each vertex of a and retaining the root of a. It is the free pre-Lie algebra on the rooted singleton, by Chapoton and Livernet [3, Theorem 1.9 and Corollary 1.10]. Define $\rho : \mathcal{U} \to \mathcal{R}$ by summing over all choices of a root. Then

$$\rho(t \diamond v) = \rho(t) \triangleright \rho(v) + \rho(v) \triangleright \rho(t). \tag{10}$$

The first summand counts joined trees whose root lies in t, and the second those whose root lies in v; the choices of root and attachment vertex agree on both sides, including multiplicities from automorphisms.

Bergeron and Loday [4, Theorem 3.1] prove that the operad morphism ComMag $\to$ preLie induced by symmetrization is injective. This gives the one-generator injection used here. Indeed, over $\mathbb{R}$, coinvariants of a finite symmetric group are identified with invariants by averaging; the inverse is the quotient map. Passing an equivariant injection to coinvariants consequently remains injective. Tensoring each arity with the one-dimensional generator space and taking the direct sum proves that the symmetric-product map from X to $\mathcal{R}$ is injective. By (10), $\rho\psi$ is precisely that map. Thus ψ is injective as well. □

For a basis unrooted tree t define the elementary potential

$$H_t(V)(q) = \sum_{c : E(t) \to \{1,\dots,d\}} \prod_{a \in \mathrm{Vert}(t)} \partial^{\alpha_a(c)} V(q), \tag{11}$$

where $E(t)$ is its edge set and $\alpha_a(c)$ is the multiindex of colors incident at vertex a. For the singleton this is V. Extend H linearly to $\mathcal{U}$. Differentiation of the finite sum increments the jet at each possible vertex, giving

$$\nabla H_t(V) \cdot \nabla H_v(V) = H_{t \diamond v}(V). \tag{12}$$

For potentials h, k, our Poisson convention gives $\{h, T\} = \nabla h \cdot p$ and $\{k, \{h, T\}\} = \nabla k \cdot \nabla h$. Induction in (4) therefore shows

$$\Phi_{d,V}(x) = H_{\psi(x)}(V) \qquad (x \in X). \tag{13}$$

Lemma 3.2. *Every nonzero finite linear combination $h \in \mathcal{U}$ is detected by $H_h(V)$ for some dimension $d \geq 1$ and some polynomial potential V.*

Proof. Choose a tree t with nonzero coefficient in h, and let r be its number of vertices. As a polynomial in the independent symmetric jet variables $Z_\alpha = \partial^\alpha V(0)$, its contraction has total jet degree r, so trees with other vertex numbers cannot cancel a degree-r monomial. For $r = 1$, the singleton is the only tree and its monomial is Z_0; use $d = 1$.

If $r \geq 2$, take $d = r - 1$ and color its edges with distinct coordinate indices. Let α_a be the resulting incidence multiindices and set

$$M_t = \prod_{a \in \mathrm{Vert}(t)} Z_{\alpha_a}.$$

Every coordinate index occurs exactly twice. A contraction of another r-vertex tree producing M_t must therefore use each color on exactly one edge. Its endpoints are the two vertices whose incidence multiindices contain that color. The multiset of incidence multiindices reconstructs

the colored graph, up to permutations of equal vertices, and hence its unrooted tree type. In the two-vertex case the two multiindices coincide; exchanging the endpoints still reconstructs the same tree. Thus only trees isomorphic to t can contribute M_t. Its coefficient in H_t is a positive integer, since the chosen coloring contributes and all terms have positive coefficients. Its coefficient in H_h is consequently nonzero.

The nonzero jet polynomial H_h depends on finitely many variables and has a nonzero real evaluation. One may prove this by induction on the number of variables: make a nonzero coefficient polynomial nonzero, then avoid the finitely many roots in the remaining variable. Arbitrary prescribed jets up to a finite order D are realized by

$$V(q) = \sum_{|\alpha| \le D} z_\alpha \frac{q^\alpha}{\alpha!}, \qquad \partial^\alpha V(0) = z_\alpha.$$

Taking D at least the largest derivative order needed gives the required polynomial potential. □

Proof of Theorem 1.1. Let K be the joint kernel over all dimensions and polynomial potentials. Each realization preserves homogeneous momentum degree. Independence of the different polynomial degrees in p makes K a graded Lie ideal. Lemmas 3.1 and 3.2, with (13), show that $K \cap X = 0$. Proposition 2.4 then gives $K = 0$. Every polynomial potential is smooth, so the joint kernel over smooth potentials is contained in K and is also zero. □

A The associative realization

We recall a short algebraic proof of the fact used in Lemma 2.1. For an ordinary Lie algebra L, choose an ordered vector-space basis (e_i) and reduce tensor words by

$$e_j e_i \longmapsto e_i e_j + [e_j, e_i] \qquad (j > i)$$

in its enveloping algebra. These reductions terminate by tensor length and, at fixed length, inversion number. Disjoint reductions commute. The only overlap is a descending three-letter word; the difference of its two reductions, after reducing shorter terms, is the Jacobi expression and is zero. Induction on the same decreasing measure resolves every sequence of reductions. Nondecreasing basis monomials therefore form a basis of the enveloping algebra, and the length-one terms give an injection $L \to U(L)$. For the free Lie algebra $L(S)$, the enveloping algebra and the free associative algebra $\mathbb{R}\langle S\rangle$ have the same universal property: a map from S to an associative algebra extends uniquely, first as a Lie map and then as an associative map. They are generator-preservingly isomorphic. Thus $L(S)$ embeds in $\mathbb{R}\langle S\rangle$. The argument also applies to infinite S, since every word and reduction is finite.

B Exact finite computations

The accompanying code reconstructs the mechanical-tree matrices and checks the degree-lowering formulas for every basis element through original-letter order 18. A potential tree with b leaves has order $2b-1$, its generator y_s has order $2b$, and A has order one. Weighted Lyndon words on these free generators, together with the binary-tree potential basis, provide the input columns. The marker is a comb with ten leaves. Sparse associative expansions and tree contractions use integer coefficients; no floating-point rank threshold is involved.

Order	6	10	14	16	18
Source dimension	5	39	320	938	2786
Mechanical-tree rank	5	39	320	938	2786

Each rank is computed modulo the prime 1000003. Full column rank modulo a prime implies full column rank of the integer matrix over $\mathbb{Q}$ and $\mathbb{R}$, since a full-size minor has determinant nonzero modulo that prime. The stored matrices have 6705 columns in 91 order/degree blocks; a separate elimination replay checks every stored pivot and its integrity hash. Rebuilding the columns and replaying them are distinct checks.

Across the 6611 positive-degree columns, the degree-one action is checked on 362 elements and the projection/retraction formula (9) on 6249 elements. All 1019 associative y-words through the cutoff pass marker decoding. An additional direct sparse-polynomial implementation in dimension two, with degree-five potentials and seeds 519 and 2026, checks all 40 columns through order eight for each potential at $q = 0$ and $p = (1, 2)$. Exact rational differentiation agrees with the tree contractions in all 80 value comparisons. These are pointwise calibration checks of the implementation. Iterated lowering through order five comprises 8 inputs and 17 steps, all ending in a nonzero potential. These finite results support implementation correctness; the all-order theorem is proved in Sections 2–3.

The following finite validation procedure takes a cutoff $2 \le N \le 18$ and a prime p_0; it chooses a marker with $\lfloor N/2 \rfloor + 1$ leaves and returns the dimension/rank table and replayable matrices and pivots.

1. Enumerate binary potential trees and weighted Lyndon words of order $1, \dots, N$; expand each Lyndon bracket by integer commutators.
2. For every positive-degree column, compute its bracket with the marker. Check word decoding in degree one and projection followed by retraction in higher degree.
3. Construct the integer mechanical-tree image of every column, group columns by order and momentum degree, and eliminate modulo p_0 using the lexicographically first available row as pivot. Reject a dependent column. Save the integer columns, pivots, labels and integrity hashes for replay.

If a block has C columns and R possible rows, dense elimination uses at most $O(RC^2)$ field operations and $O(RC)$ storage; the implementation uses sparse dictionaries. These counts concern elimination of an already formed matrix. Tree generation, word expansion and coefficient bit growth are additional costs, and no polynomial bound in the order cutoff is asserted. Caches for sparse intermediate mappings retain at most 4096 entries per function; eviction changes neither the arithmetic nor the result.

Reproduction commands, the tested library, the complete order table and small validation summaries are supplied with the manuscript. The drivers save full local matrices, lowering traces and source/output hashes in fresh run directories. The raw matrices can be regenerated from the code.

References

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Statement on AI assistance

OpenAI ChatGPT assisted with literature exploration and the initial proof development. OpenAI Codex assisted with source verification, proof checking, manuscript preparation, and implementation and testing of the computational material. These activities are AI-assisted checks; they do not constitute independent human peer review. The manuscript is submitted under the author's name, and independent mathematical review is requested.